CBCS SCHEME

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BCS302

Third Semester B.E./B.Tech. Degree Examination, June/July 2025
Digital Design and Computer Organization

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Demonstrate the positive and negative logic using AND gate.	05	L2	CO1
	b.	Show that NAND and NOR function are commutative and but not associative.	05	L3	CO1
	c.	Simplify the Boolean function F(w,x,y,z) = Σm (0,1,2,4,6,7,9,12,14) using K-Map and implement using NAND gates.	10	L3	CO1
		OR			
Q.2	a.	What is Binary Logic? List out any 4 laws of logic.	05	L1	CO1
	b.	Find the POS expression for F(A.B.C,D) = ∏M(2,3,5,8,10,13,14) + d(1,6,7,11) and realize it using NOR gates.	05	L3	CO1
	c.	Simplify the following Boolean functions using K-map and write the verilog program for realizing the minimized expression. i) F(X,Y,Z) = Σm (0,1,4,5,6) +d (2,3) ii) F(W,X,Y,Z) = Σm (5,6,7,12,14,15) +d(9,11,13)	10	L3	CO1
		Module – 2			
Q.3	a.	Define Decoder Implement the following boolean functions using a decoder $F_1(A,B,C) = \Sigma m(1,3,4,7)$ $F_2(A,B,C) = \Sigma m(0,2,,3,6)$	06	L3	CO2
	b.	Write the verilog program to implement full adder and full sub tractor circuits.	06	L3	CO2
	c.	Design an octal to Binary Encoder.	08	L3	CO2
		OR			
Q.4	a.	Define Multiplexer. Implement the Boolean function F (A,B,C,D) = Σm(1,3,4,11,12,13,14,15) with 8:1 multiplexer.	06	L3	CO2
	b.	Explain the working of 4 – bit adder using 4 full adders.	06	L2	CO2
	c.	Write the characteristic equation, Excitation table and FSM representations for SR,JK and D flip flops	08	L2	CO2
		Module – 3			
Q.5	a.	With a block diagram explain the processor and Memory communication.	06	L2	CO3
	b.	With relevant examples, Explain the following addressing modes. i) Index ii) Base with index and offset iii) Indirect.	06	L2	CO3
	c.	Demonstrate the instruction execution and sequencing for C←[A] +[B]	08	L2	CO3

		OR			
Q.6	a.	Describe the Big Endian and little endian address assignment with examples	06	L2	CO3
	b.	The Registers R1 and R2 has decimal values 1200 and 4600. Calculate the EA of the memory operand in each of the following instructions when they are executed in sequence. i) load 20 (R1), R5 ii) Move # 3000, R5 iii) Store R5, 30(R1,R2) iv) add – (R2), R5 v) Sub (R1)+, R5 vi) add (R2)+,R1	06	L3	CO3
	c.	Demonstrate the Branching operations using a loop to add n – numbers with block diagram.	08	L3	CO3
		Module – 4			
Q.7	a.	Explain the effect of size, cost and speed in Memory hierarchy.	10	L2	CO4
	b.	Explain Centralized and distributed Bus arbitration approaches.	10	L2	CO4
		OR			
Q.8	a.	Describe the different memory mapping functions.	10	L2	CO4
	b.	Explain how to handle interrupt from multiple devices using daisy chain and priority scheme.	10	L2	CO4
		Module – 5			
Q.9	a.	With a neat diagram, explain the single bus organization of the data path inside a processor.	10	L2	CO5
	b.	Describe in detail, the basic idea of instruction pipeline.	10	L2	CO5
		OR			
Q.10	a.	Explain the complete set of operations involved in executing the instruction ADD (R3), R1 along with control sequence.	10	L2	CO5
	b.	Explain the process of storing a word from processor to memory.	10	L2	CO1

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